# Report about the Model Selection and the Project

## Introduction:

This project is about a CKD(Chronic Kidney Disease) dataset. The dataset has patient features that represent values of different tests taken for patients. These patients features represent the input variables. The Project is to find the best-fitting model that would predict whether a patient has CKD or not depending on the inputs given. The target variable in this dataset is a categorical column indicating whether the patient has CKD or not. This dataset has nearly 26 columns and the idea behind this project is to reduce the number of input columns as much as possible without compromising the model's accuracy of predicting CKD for a patient.

## Steps followed to achieve at the Best-Fitting Model:

1. CKD dataset was loaded and checked for missing values, incorrect values, unwanted strings, outliers (lesser/higher), correct datatype based on its data and saved a pre-processed CKD dataset.
2. This pre-processed CKD dataset was analysed with Univariate Analysis and Bi-variate Analysis concepts to understand the distribution/measure of various features in the dataset.
3. **Using SelectKBest feature-selection, checked with various 'k' values for seven Classification algorithms starting from Logistics, SVML, SVMNL, KNN, Navie, DecisionTree and RandomForest.**
4. SelectKBest Feature selection helped to find the best k value for this dataset. And applying different k values in all the seven classification algorithms helped to find out the best-fitting Model for this dataset.
5. **The best 'k' value was found to be '8' for RandomForest Algorithm with Accuracy of 100%; RandomForest Algorithm's accuracy was the highest consistently for all the tried k values, starting from 3 through 10. So, have chosen RandomForest Algorithm with best 'k' value to be 8, for this CKD dataset.**
6. **Created a RandomForest Model and deployed it.** This Model will get 8 inputs from the user to predict whether the patient has CKD or not. The inputs needed are age, al, bgr, bu,sc, hemo, pcv and wc.

'K' values and Accuracy of different Models – SelectKBest Feature Selection Results: (see below)

| S.No | K value | Logistic | SVML | SVMNL | KNN | Navie | DT | RF | Comments |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 3 | 0.84 | 0.83 | 0.84 | 0.82 | 0.84 | 0.86 | 0.9 | RF |
| 2 | 4 | 0.9 | 0.88 | 0.89 | 0.85 | 0.84 | 0.95 | 0.95 | RF |
| 3 | 5 | 0.91 | 0.94 | 0.94 | 0.93 | 0.9 | 0.97 | 0.98 | RF |
| 4 | 6 | 0.96 | 0.97 | 0.98 | 0.95 | 0.95 | 0.96 | 0.99 | RF |
| 5 | 7 | 0.97 | 0.98 | 0.98 | 0.98 | 0.96 | 0.98 | 0.99 | RF |
| 6 | 8 | 0.97 | 0.97 | 0.98 | 0.97 | 0.96 | 0.97 | 1 | RF |
| 7 | 9 | 0.97 | 0.98 | 0.98 | 0.97 | 0.96 | 0.97 | 0.96 | SVM |
| 8 | 10 | 0.97 | 0.98 | 0.98 | 0.97 | 0.94 | 0.97 | 0.98 | RF, SVM |

## Conclusion:

As we can see from the above table of results derived from SelectKBest Feature Selection, throughout the trials right from the 'k' value = 3 through 'k' value = 10, RandomForest algorithm has been consistently showing highest accuracy values. Also, if we could note, the Accuracy score is 100% for RandomForest for 'k' value = 8. So, it is ideal to create a RandomForest model with 'k' value = 8.